

EXPERIMENT - 40

1. Problem Statement

Write a menu-driven C program that reads a student's first name, middle name and last name. Use separate user-defined functions for each operation. Use call by reference by passing pointers wherever a function modifies a string or returns multiple results. Perform the following tasks:

- Combine the first, middle and last names to form and display the student's full name.
- Find and display the length of the full name.
- Copy the full name into another string and display the copied string.
- Compare the first name and last name and display whether they are equal or different.
- Display the full name in reverse order, that is, last name, middle name and first name.
- Count and display the number of vowels, consonants, digits and spaces in the full name.
- Read a word and check whether it occurs in the full name.
- Exit.

Header files used: `stdio.h`, `string.h` and `ctype.h`.

2. Algorithm

Main Program

Step 1: START

Step 2: Read `firstName`, `middleName` and `lastName`.

Step 3: Call `combineNames()` once so that `fullName` is ready for the other options.

Step 4: Display the menu and read choice.

Step 5: Using a switch statement on choice:

 If choice is 1, call `combineNames()` and display `fullName`.

 If choice is 2, call `findLength()` and display length.

 If choice is 3, call `copyName()` and display `copiedName`.

 If choice is 4, call `compareNames()` and display equal if `comparisonResult` is 0, otherwise different.

 If choice is 5, call `reverseNameOrder()` and display `reverseName`.

 If choice is 6, call `countCharacters()` and display the four counts.

 If choice is 7, read word, call `searchWord()` and display present if `searchResult` is 1, otherwise not present.

 If choice is 8, display "Exiting the program."

Step 6: Repeat from Step 4 while choice is not equal to 8.

Step 7: STOP

Function `combineNames()`

Step 1: Copy `firstName` into `fullName` using `strcpy()`.

Step 2: Append a space and then `middleName` to `fullName` using `strcat()`.

Step 3: Append a space and then lastName to fullName using strcat().

Step 4: Return to the calling function.

Function findLength()

Step 1: Set the value at length to strlen(fullName).

Step 2: Return to the calling function.

Function copyName()

Step 1: Copy source into destination using strcpy().

Step 2: Return to the calling function.

Function compareNames()

Step 1: Set the value at comparisonResult to strcmp(firstName, lastName).

Step 2: Return to the calling function.

Function reverseNameOrder()

Step 1: Copy lastName into reverseName using strcpy().

Step 2: Append a space and then middleName using strcat().

Step 3: Append a space and then firstName using strcat().

Step 4: Return to the calling function.

Function countCharacters()

Step 1: Set the values at vowels, consonants, digits and spaces to 0.

Step 2: Set len = strlen(fullName) and repeat for i = 0 to len – 1:

Set ch = tolower(fullName[i]).

If isalpha(ch) is true, add 1 to vowels when ch is 'a', 'e', 'i', 'o' or 'u', and add 1 to consonants otherwise.

Otherwise, if isdigit(ch) is true, add 1 to digits.

Otherwise, if isspace(ch) is true, add 1 to spaces.

Step 3: Return to the calling function.

Function searchWord()

Step 1: If strstr(fullName, word) is not NULL, set the value at searchResult to 1, otherwise set it to 0.

Step 2: Return to the calling function.

3. Test Case Table

#	Operation	Input	Expected Output	Actual Output	Result
1	1. Combine	Rahul, Kumar, Sharma	Rahul Kumar Sharma, with single spaces	Full name: Rahul Kumar Sharma	PASS
2	2. Length	the same name	18, counting the two spaces	18, so the spaces are in there	PASS
3	3. Copy	the same name	An identical copy in a second string	Copy came out the same, original untouched	PASS
4	4. Compare	Rahul against Sharma	Different	Different, strcmp gave a negative number	PASS
5	5. Reverse order	the same name	Sharma Kumar Rahul	Sharma Kumar Rahul, the letters not reversed	PASS
6	6. Count	the same name	6 vowels, 10 consonants, 0 digits, 2 spaces	6, 10, 0 and 2, adding up to 18	PASS
7	7. Search	the word Kumar	Present	Said it is present	PASS
8	7. Search	the word Singh	Not present	Said it is not present	PASS
9	4. Compare	Rahul, Kumar2, Rahul	Equal, since both ends are the same word	Said they are equal	PASS
10	6. Count	the same name	6 vowels, 9 consonants, 1 digit, 2 spaces	The 2 was counted as a digit, not a consonant	PASS
11	8. Exit	choice = 8	Exiting the program, and the loop ends	Loop ended	PASS

4. Source Code

names40.c

```
/* Experiment 40. The three parts of the name are read once at the start
and every option works on those same three.
The question lists some of the prototypes without a star, like
size_t length and int vowels, but the line above them asks for call
by reference wherever a function sends back more than one result.
Both cannot be true at the same time, so I went with the pointers.
Without them the function changes its own copy, main never sees it,
and option 6 would print whatever junk was lying in memory. */

#include <stdio.h>
#include <string.h>
#include <ctype.h>

void combineNames(const char firstName[], const char middleName[],
                 const char lastName[], char fullName[]);
void findLength(const char fullName[], size_t *length);
void copyName(const char source[], char destination[]);
void compareNames(const char firstName[], const char lastName[],
                 int *comparisonResult);
void reverseNameOrder(const char firstName[], const char middleName[],
                     const char lastName[], char reverseName[]);
void countCharacters(const char fullName[], int *vowels, int *consonants,
                    int *digits, int *spaces);
void searchWord(const char fullName[], const char word[], int *searchResult);

int main(void)
{
    char firstName[50], middleName[50], lastName[50];
    char fullName[160], copiedName[160], reverseName[160], word[50];
    size_t length;
    int comparisonResult, searchResult;
    int vowels, consonants, digits, spaces;
    int choice;

    printf("Enter the first name: ");
    scanf("%s", firstName);
    printf("Enter the middle name: ");
    scanf("%s", middleName);
    printf("Enter the last name: ");
    scanf("%s", lastName);

    /* fullName is built here as well, not only inside option 1. Four of
    the options read it, and nothing stops the user from picking
    option 2 first, which on my first run printed a length of 0. */
    combineNames(firstName, middleName, lastName, fullName);

    do {
        printf("\nMENU\n");
        printf("1. Combine the names and display the full name\n");
        printf("2. Display the length of the full name\n");
        printf("3. Copy the full name and display the copy\n");
        printf("4. Compare the first name and the last name\n");
        printf("5. Display the name in reverse order\n");
        printf("6. Count the vowels, consonants, digits and spaces\n");
        printf("7. Search for a word in the full name\n");
        printf("8. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                combineNames(firstName, middleName, lastName, fullName);
                printf("Full name: %s\n", fullName);
                break;

            case 2:
                findLength(fullName, &length);
```

```

        printf("Length of the full name: %zu\n", length);
        break;

    case 3:
        copyName(fullName, copiedName);
        printf("Copied name: %s\n", copiedName);
        break;

    case 4:
        compareNames(firstName, lastName, &comparisonResult);
        if (comparisonResult == 0)
            printf("The first name and the last name are equal\n");
        else
            printf("The first name and the last name are different\n");
        break;

    case 5:
        reverseNameOrder(firstName, middleName, lastName, reverseName);
        printf("Name in reverse order: %s\n", reverseName);
        break;

    case 6:
        countCharacters(fullName, &vowels, &consonants, &digits, &spaces);
        printf("Vowels: %d\n", vowels);
        printf("Consonants: %d\n", consonants);
        printf("Digits: %d\n", digits);
        printf("Spaces: %d\n", spaces);
        break;

    case 7:
        printf("Enter the word to search: ");
        scanf("%s", word);
        searchWord(fullName, word, &searchResult);
        if (searchResult)
            printf("The word %s is present in the full name\n", word);
        else
            printf("The word %s is not present in the full name\n", word);
        break;

    case 8:
        printf("Exiting the program.\n");
        break;
    }
} while (choice != 8);

return 0;
}

/* strcpy goes first because it wipes whatever was in fullName and
starts clean, and every strcat after it adds on to the end. The " "
lines are not decoration. I left them out at first and got
RahulKumarSharma, which option 2 then measured as 16 instead of
18. */
void combineNames(const char firstName[], const char middleName[],
                 const char lastName[], char fullName[])
{
    strcpy(fullName, firstName);
    strcat(fullName, " ");
    strcat(fullName, middleName);
    strcat(fullName, " ");
    strcat(fullName, lastName);
}

/* The whole function is one line, and the only interesting part is the
star. *length means "the variable whose address I was handed", so
this writes straight into main's own length. */
void findLength(const char fullName[], size_t *length)
{
    *length = strlen(fullName);
}

/* destination is the only one of the two that is not const, because it
is the one being written into. strcpy carries the '\0' across too,
and that is what makes the result a real string rather than a pile

```

```

    of loose characters. */
void copyName(const char source[], char destination[])
{
    strcpy(destination, source);
}

/* strcmp does not say yes or no. It gives 0 when the two strings are
the same, and a positive or negative number otherwise depending on
which one would come first in a dictionary. 0 meaning equal felt
backwards to me at first. The number goes back through the pointer
and main decides what to print. */
void compareNames(const char firstName[], const char lastName[],
                  int *comparisonResult)
{
    *comparisonResult = strcmp(firstName, lastName);
}

/* Same idea as combineNames, only the last name goes in first. The
question suggests strtok, but I did not need it. strtok is for
cutting one long string into pieces, and here the three pieces
arrive already separate, so joining them up just to cut them apart
again would be work for nothing. */
void reverseNameOrder(const char firstName[], const char middleName[],
                     const char lastName[], char reverseName[])
{
    strcpy(reverseName, lastName);
    strcat(reverseName, " ");
    strcat(reverseName, middleName);
    strcat(reverseName, " ");
    strcat(reverseName, firstName);
}

/* Four answers come out of here and a function can only return one
value, so this is where the pointers really earn their place.
The four lines setting them to 0 are not there for neatness. I had
left them out, the first call was perfect, and the second call on
the same name told me there were 12 vowels, because it had carried
on adding to the old counts.
tolower first means the vowel test only has to list the five small
letters, and the else if chain makes sure a character lands in
exactly one of the four counts. */
void countCharacters(const char fullName[], int *vowels, int *consonants,
                    int *digits, int *spaces)
{
    size_t i, len;
    char ch;

    *vowels = 0;
    *consonants = 0;
    *digits = 0;
    *spaces = 0;

    len = strlen(fullName);

    for (i = 0; i < len; i++) {
        ch = tolower((unsigned char) fullName[i]);

        if (isalpha((unsigned char) ch)) {
            if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u')
                (*vowels)++;
            else
                (*consonants)++;
        } else if (isdigit((unsigned char) ch)) {
            (*digits)++;
        } else if (isspace((unsigned char) ch)) {
            (*spaces)++;
        }
    }
}

/* strstr does not answer yes or no either. It hands back the address
inside fullName where the word begins, or NULL if it is not there.
main only wants a yes or a no, so that address is turned into 1 or 0
here and passed back through the pointer. */

```

```
void searchWord(const char fullName[], const char word[], int *searchResult)
{
    if (strstr(fullName, word) != NULL)
        *searchResult = 1;
    else
        *searchResult = 0;
}
```

5. Compilation and Output

I compiled it with `gcc -Wall -Wextra names40.c -o names40`. Those flags bring out the small warnings as well, and it built clean.

Reading the Name, Then Combining and Measuring It

```
adilshakil@ADILs-MacBook-Air MyAssignment % gcc -Wall -Wextra names40.c -o names40
adilshakil@ADILs-MacBook-Air MyAssignment % ./names40
Enter the first name: Rahul
Enter the middle name: Kumar
Enter the last name: Sharma

MENU
1. Combine the names and display the full name
2. Display the length of the full name
3. Copy the full name and display the copy
4. Compare the first name and the last name
5. Display the name in reverse order
6. Count the vowels, consonants, digits and spaces
7. Search for a word in the full name
8. Exit
Enter your choice: 1
Full name: Rahul Kumar Sharma

MENU
1. Combine the names and display the full name
2. Display the length of the full name
3. Copy the full name and display the copy
4. Compare the first name and the last name
5. Display the name in reverse order
6. Count the vowels, consonants, digits and spaces
7. Search for a word in the full name
8. Exit
Enter your choice: 2
Length of the full name: 18
```

Copying and Comparing

```
adilshakil@ADILs-MacBook-Air MyAssignment % gcc -Wall -Wextra names40.c -o names40
adilshakil@ADILs-MacBook-Air MyAssignment % ./names40
Enter the first name: Rahul
Enter the middle name: Kumar
Enter the last name: Sharma

MENU
1. Combine the names and display the full name
2. Display the length of the full name
3. Copy the full name and display the copy
4. Compare the first name and the last name
5. Display the name in reverse order
6. Count the vowels, consonants, digits and spaces
7. Search for a word in the full name
8. Exit
Enter your choice: 3
Copied name: Rahul Kumar Sharma

MENU
1. Combine the names and display the full name
2. Display the length of the full name
3. Copy the full name and display the copy
4. Compare the first name and the last name
5. Display the name in reverse order
6. Count the vowels, consonants, digits and spaces
7. Search for a word in the full name
8. Exit
Enter your choice: 4
The first name and the last name are different
```

Reverse Order and the Character Counts

```
adilshakil --zsh -- 159x31

MENU
1. Combine the names and display the full name
2. Display the length of the full name
3. Copy the full name and display the copy
4. Compare the first name and the last name
5. Display the name in reverse order
6. Count the vowels, consonants, digits and spaces
7. Search for a word in the full name
8. Exit
Enter your choice: 5
Name in reverse order: Sharma Kumar Rahul

MENU
1. Combine the names and display the full name
2. Display the length of the full name
3. Copy the full name and display the copy
4. Compare the first name and the last name
5. Display the name in reverse order
6. Count the vowels, consonants, digits and spaces
7. Search for a word in the full name
8. Exit
Enter your choice: 6
Vowels: 6
Consonants: 10
Digits: 0
Spaces: 2
█
```

Searching for a Word That Is There and One That Is Not

```
adilshakil --zsh -- 159x31

MENU
1. Combine the names and display the full name
2. Display the length of the full name
3. Copy the full name and display the copy
4. Compare the first name and the last name
5. Display the name in reverse order
6. Count the vowels, consonants, digits and spaces
7. Search for a word in the full name
8. Exit
Enter your choice: 7
Enter the word to search: Kumar
The word Kumar is present in the full name

MENU
1. Combine the names and display the full name
2. Display the length of the full name
3. Copy the full name and display the copy
4. Compare the first name and the last name
5. Display the name in reverse order
6. Count the vowels, consonants, digits and spaces
7. Search for a word in the full name
8. Exit
Enter your choice: 7
Enter the word to search: Singh
The word Singh is not present in the full name
█
```

A Second Name Where the First and Last Are the Same

```
adilshakil — -zsh — 159x31

MENU
1. Combine the names and display the full name
2. Display the length of the full name
3. Copy the full name and display the copy
4. Compare the first name and the last name
5. Display the name in reverse order
6. Count the vowels, consonants, digits and spaces
7. Search for a word in the full name
8. Exit
Enter your choice: 8
Exiting the program.
adilshakil@ADILs-MacBook-Air MyAssignment % ./names40
Enter the first name: Rahul
Enter the middle name: Kumar2
Enter the last name: Rahul

MENU
1. Combine the names and display the full name
2. Display the length of the full name
3. Copy the full name and display the copy
4. Compare the first name and the last name
5. Display the name in reverse order
6. Count the vowels, consonants, digits and spaces
7. Search for a word in the full name
8. Exit
Enter your choice: 4
The first name and the last name are equal
█
```

The Digit Counter, and Exit

```
adilshakil — -zsh — 159x31

MENU
1. Combine the names and display the full name
2. Display the length of the full name
3. Copy the full name and display the copy
4. Compare the first name and the last name
5. Display the name in reverse order
6. Count the vowels, consonants, digits and spaces
7. Search for a word in the full name
8. Exit
Enter your choice: 6
Vowels: 6
Consonants: 9
Digits: 1
Spaces: 2

MENU
1. Combine the names and display the full name
2. Display the length of the full name
3. Copy the full name and display the copy
4. Compare the first name and the last name
5. Display the name in reverse order
6. Count the vowels, consonants, digits and spaces
7. Search for a word in the full name
8. Exit
Enter your choice: 8
Exiting the program.
adilshakil@ADILs-MacBook-Air MyAssignment %
█
```

All eleven cases came out right.

Before anything else, something about the prototypes, because the ones in my program do not look exactly like the ones in the question. The question lists `void findLength(const char fullName[], size_t length)` and `void countCharacters(const char fullName[], int vowels, int consonants, int digits, int spaces)`, but the line above them says to use call by reference by passing pointers wherever a function returns multiple results. Those two things cannot both be done. Written the way they are printed, the function receives a copy of the variable, changes the copy, and main is none the wiser. So I wrote them as `size_t *length` and `int *vowels` and so on, and main passes `&length` and `&vowels`. `comparisonResult` and `searchResult` are the same. The three that fill in a string, `combineNames`, `copyName` and `reverseNameOrder`, did not need an extra star, because passing an array name already passes its address, so what they write is visible in main by itself.

The bug that cost me the most time was in `countCharacters`, and it was a sneaky one, because the first call was perfect. I ran option 6 a second time on the same name and it told me there were 12 vowels and 20 consonants. I had not set the four counters to 0 at the top of the function, so they still held the previous answer and the loop simply carried on adding to it. Four lines at the start and it was fine.

The easier one was in `combineNames`. My first version was three `strcat` calls and nothing else, so Rahul Kumar Sharma came out as `RahulKumarSharma`. I only spotted it through option 2, which reported 16 characters when I was expecting 18, and the two missing ones were the spaces.

I did not use `strtok` anywhere, even though the question offers it for option 5. `strtok` is a tool for cutting one long string into pieces, and in `reverseNameOrder` the three pieces are already sitting in separate arrays. Joining them into a full name only to chop it back up would be work for nothing, so `strcpy` for the last name and `strcat` for the other two does the job.

The second run has a digit sitting in the middle name on purpose. A real name has no digits in it, so the digit counter would have shown 0 for ever and I would never have found out whether that branch worked at all. With `Kumar2` it reports 1 digit and 9 consonants instead of 10, which is right, because `isalpha()` turns the 2 away before the consonant branch gets a look at it.

One last thing worth saying out loud, since the word `reverse` is in the question. Option 5 reverses the order of the three names, not the letters, so `Rahul Kumar Sharma` becomes `Sharma Kumar Rahul` and not `amrahS ramuK luhaR`.